

Namanh Bui Vu

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Education

Technische Universität Hamburg (TUHH)

Bachelor of Science in Computer Science · Algorithms & Data Structures, Databases, Networks, C/C++

Hamburg, Germany

Oct 2024 – Expected 2028

National University of Singapore (NUS)

Incoming Exchange Semester, Computer Science · Software Engineering, Software Testing, ML, AI

Singapore

Jan 2027 – May 2027

Experience

Arbio Group GmbH

Software Engineer (Product Engineering) · TypeScript, NestJS, AWS, PostgreSQL

Berlin, Germany

May 2026 – Present

- Cut guest lock-out incidents from **50+ a month to under 10** by rebuilding how Arbio's door codes reach its locks, in **TypeScript on AWS** – now live on **434 locks** across 1,000+ apartments, with an automatic fallback when a vendor fails.
- Made it reliable against two vendor **REST APIs** that confirm late and offer no safe retry: one **PostgreSQL** writer per reservation, lease-based job claiming, and compare-and-swap with rollback, so a lost response never leaves a duplicate or dead code. **1,500+ tests**.
- Replaced manual reading of owner documents with an **8-stage AI pipeline** (Anthropic API) citing the exact page behind every value, cutting **166 hand-checked data points** per property and lifting extraction accuracy from **0.42 to 0.81 F1**.
- Worked in a team of five engineers who reviewed my **186 pull requests**, took their direction across **637 review threads** and closed 561, and dropped my own service design after the infrastructure lead disagreed with it.

Position One GmbH

AI Engineer (promoted from intern) · Python, LangGraph, LangChain, Langfuse, PostgreSQL

Hamburg, Germany

Sep 2025 – Apr 2026

- Cut **15+ hours of manual work per week** with LLM agent pipelines in **Python** (LangGraph, LangChain) across sales, operations, and e-commerce – instrumented with **Langfuse tracing and evaluations** so quality was measured, not assumed.
- Built a store generator that keeps **1,000+ products** in sync with supplier **APIs**: hash-based change detection over a **PostgreSQL** store, **multithreaded** image ingestion into object storage, and backoff against two rate-limited endpoints.

Technische Universität Hamburg

Teaching Assistant · C, Python

Hamburg, Germany

Oct 2025 – Jan 2026

- Taught **25+ first-semester** Computer Science and Data Science students to program in **C and Python**, going through their code with them each week and explaining what was wrong in plain terms.

Projects

Minus One – Six-Stem Song Separation

Next.js 16, TypeScript, Demucs, RunPod serverless GPU, Web Audio API, Redis

Live Demo

- Splits any uploaded track into **six instrument stems** (Demucs `htdemucs_6s`) on a queue-based **serverless GPU worker**, behind a provider interface with a second inference backend as fallback; blob storage, Redis rate limits, TTL cleanup jobs.
- Browser mixer plays all six stems **in sync** through a Web Audio gain graph, correcting drift above 40 ms every 500 ms; 13 unit suites plus Playwright end-to-end flows.

Football Match Prediction – Beating Bookmaker Odds

Python, CatBoost, scikit-learn, SHAP, pandas, pytest

GitHub

- **Backtested a >30% ROI edge over 8 seasons against closing bookmaker odds**, using leakage-safe walk-forward validation, SHAP feature selection, and probability calibration so the edge held out of sample.

Competitions

- **AdventureX 2026, Hangzhou – 1st Place, Pico Track.** Selected for a **fully funded seat** at China's largest hackathon (800 participants); built an immersive XR experience on ByteDance Pico hardware in Unity/C#.
- **{Tech: Europe} Berlin – 1st Place Arbio Track, 3rd Overall** with Airbn (GPT-4o, OpenCV, FastAPI), **the win that converted directly into the Arbio engineering role** · **Cursor AI – 1st Place, Google DeepMind Gemini Track** (400+ participants) · **CodeRabbit × Windsurf – 1st Place Overall, \$2,000** (110 participants).

Content & Community

- **Co-founded “FAANG/MANGO+ für Deutsche”**, the first German-language Big-Tech career community (**300+ members**), helping students land internships and working-student roles.
- **Tech creator.** Grew and sold several social-media accounts (50k+ followers, 20M+ views); now publishes software-career content at **@namb.tech – 460k+ views**.

Technical Skills

Languages: Python, TypeScript/JavaScript, C, C++, SQL, Haskell | **Spoken:** German (native), English (fluent)

Backend & Cloud: NestJS, FastAPI, Node.js, REST APIs, PostgreSQL, Docker, AWS, CI/CD | **Frontend:** React, Next.js, Tailwind

AI / ML: Anthropic, OpenAI & Gemini APIs, LangGraph, Langfuse, RAG, scikit-learn